Qiutian(Autumn) Wang

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EDUCATION

McGill University, Montreal, Canada 2025 – Present
Master of Applied Science, Translational Biomedical Engineering | CGPA: 3.91/4.00
Relevant Coursework (in progress): Applied Machine Learning (PyTorch, CNNs, Transformers, Model Evaluation)

John Abbott College, Montreal, Canada 2024 – 2025
AEC in Full-Stack Development | Dean's List 2024

University of York, York, United Kingdom 2023 – 2025
Master of Science, Computer Science (Online Part-time)

Concordia University, Montreal, Canada 2019 – 2022
Bachelor of Science, Honours Specialization in Cell and Molecular Biology | Dean's List 2019-2020, MCCUC Scholarship 2021

PROJECTS

DirectorMind: AI Co-Director | Python, JavaScript, NVIDIA Nemotron, Meterless, Zetaris Oct 2026
Open Agent Hackathon 2026

- Designed and developed an **end-to-end agentic AI application** for filmmaking pre-production, orchestrating character analysis, story ideation, script reasoning, and storyboard generation with continuity reviews.
- Built **multi-stage AI agent workflows** integrating **NVIDIA Nemotron via Meterless Nightshift**, enabling context-aware reasoning, evidence-based feedback, and structured creative decision support.
- Implemented **tool-augmented reasoning with Zetaris** storyboard retrieval and **H-MEM-inspired persistent character memory**, supporting contextual retrieval, prompt reinjection, and automated integration testing.

Breast Ultrasound Lesion Segmentation | PyTorch, U-Net Sept 2026 - Oct 2026

- Built a reproducible end-to-end breast ultrasound lesion segmentation pipeline with PyTorch, including training, evaluation, threshold tuning, inference, and error analysis.
- Improved validation Dice from **0.54 to 0.70** by upgrading a lightweight U-Net to a deeper architecture with BatchNorm; achieved **0.66 mean / 0.80 median Dice** on an untouched test set.
- Conducted controlled augmentation and architecture ablations, identifying model capacity as the primary performance bottleneck.

WORK EXPERIENCE

Docusys Corporation / Mansfield Medical Device Distributors, Montreal, Canada Apr 2026 – Present
Medical Device Quality Assurance / Regulatory Affairs Intern → Quality Assurance Coordinator

- Built an **AI-assisted FDA cybersecurity submission validator** for **39 test cases**, combining automated first-pass review, criterion-level traceability, and multimodal evidence checking.
- Applied **AI automation to medical-device QA/RA workflows**, including audit reporting and regulatory evidence review across **FDA, ISO 13485, Health Canada, MDSAP, QMS, and post-market surveillance**.

Yulcom Technologies, Montreal, Canada Feb 2025 – Aug 2025
Full Stack Developer

- Redesigned a **PostgreSQL medical-records database**, reducing schema complexity by **75% (~200 → ~50 tables)**.
- Built secure REST APIs and frontend features using **Django REST Framework** and **React/TypeScript**.
- Led a **4-developer team** through Agile sprints and collaborated with the AI team on chatbot API integration.

Hallett Lab, Concordia University, Montreal, Canada 2020 – 2021
Undergraduate Researcher, Supervisor: Dr. Michael Hallett (CRC Tier 1 in Bioinformatics Algorithms)

- Analyzed and visualized **transcriptomic data in R (ggplot2, dplyr)** to identify gene-expression patterns selected for downstream **DROP-seq single-cell sequencing** validation.
- Investigated drug-tolerance mechanisms in *Candida albicans* using **CRISPR-Cas9** gene editing.

SKILLS

Programming & Data: Python, SQL, R, pandas, TypeScript/JavaScript, Java
Machine Learning & Computer Vision: PyTorch, scikit-learn, NumPy, LLM Integration, AI Agent Workflows, Prompt Engineering, Image Segmentation, Model Evaluation, nilearn
Software & Cloud: Django REST Framework, React, REST APIs, Docker, AWS (EC2, S3), Azure, Git/GitHub, Automated Testing
AI Development Tools: Cursor, OpenAI Codex CLI
Medical Device & Regulatory: FDA Submissions, ISO 13485, Health Canada, MDSAP, QMS, Internal Audits, Post-Market Surveillance, CAPA, GMP, NHN
Languages: Mandarin (Native), English (Full Professional Proficiency), French (Full Professional Proficiency)